

Vector Spaces (Ch1)

Subspaces, span, sums

Vector-space facts: $-(-u) = u$, $0_{\mathbb{R}}u = 0_V$, $(-1)u = -u$, $c0_V = 0_V$, $u - v = u + (-v)$.

Subspace test: $W \leq V \iff W \neq \emptyset$ and $u, v \in W \Rightarrow u + v \in W$, $c \in \mathbb{F}$, $u \in W \Rightarrow cu \in W$.

One-line test: $W \leq V \iff W \neq \emptyset$ and $cu + v \in W$ for all $u, v \in W$, $c \in \mathbb{F}$.

Fast disproof: $0 \notin W \Rightarrow W \not\leq V$. If $W \leq V$, then $0 \in W$, $u \in W \Rightarrow -u \in W$, $u, v \in W \Rightarrow u - v \in W$.

Kernel trick: $T : V \rightarrow U$ linear $\Rightarrow \text{Ker}(T) = \{v : T(v) = 0\} \leq V$. Homogeneous linear conditions usually define subspaces.

Span: $\text{span}(S)$ = set of all finite linear combinations of vectors in S ; $\text{span}(\emptyset) = \{0\}$. Smallest-subspace property: $S \subseteq W \leq V \Rightarrow \text{span}(S) \subseteq W$.

Span identities: $A \subseteq B \Rightarrow \text{span}(A) \subseteq \text{span}(B)$; $\text{span}(\text{span}(A)) = \text{span}(A)$; $\text{span}(A \cup B) = \text{span}(A) + \text{span}(B)$.

Subspace ops: If $U, W \leq V$, then $U \cap W \leq V$ and $U + W = \text{span}(U \cup W) \leq V$.

Union criterion: $W_1 \cup W_2 \leq V \iff (W_1 \subseteq W_2) \text{ or } (W_2 \subseteq W_1)$.

Affine/coset test: $\{\sum_i \lambda_i a_i + b\} \leq V \iff b \in \text{span}\{a_i\}$. If $W \leq V$, then $w_0 + W \leq V \iff w_0 \in W$ (then $w_0 + W = W$).

Direct sums

Definition: $V_1 \oplus \dots \oplus V_n$ means every $v \in V_1 + \dots + V_n$ has a unique decomposition $v = v_1 + \dots + v_n$ with $v_i \in V_i$.

Zero-sum test: $V_1 + \dots + V_n$ is direct $\iff v_1 + \dots + v_n = 0$, $v_i \in V_i \Rightarrow v_1 = \dots = v_n = 0$.

2-subspace test: $V = U \oplus W \iff V = U + W$ and $U \cap W = \{0\}$.

Intersection form: $V_1 + \dots + V_n$ direct $\iff V_i \cap (\sum_{j \neq i} V_j) = \{0\}$ for all i .

Warning: Pairwise trivial intersections are not enough when $n \geq 3$.

LI link: If S is LI and $S = A \sqcup B$, then $\text{span}(S) = \text{span}(A) \oplus \text{span}(B)$.

Sum-map view: $\phi : V_1 \times \dots \times V_n \rightarrow V$, $\phi(v_1, \dots, v_n) = \sum_i v_i$. Then $\text{im}(\phi) = \sum_i V_i$ and the sum is direct $\iff \text{Ker}(\phi) = \{0\}$.

Finite-Dimensional Spaces (Ch2)

LI, basis, coordinates

LI: S is LI $\iff \lambda_1 v_1 + \dots + \lambda_k v_k = 0$ (distinct $v_i \in S$) implies $\lambda_1 = \dots = \lambda_k = 0$.

Dependence criterion: S dependent $\iff \exists v \in S$ with $v \in \text{span}(S \setminus \{v\})$. In particular, $0 \in S \Rightarrow S$ dependent.

Quick facts: $\emptyset$ is LI; $\{v\}$ is LI $\iff v \neq 0$; LI $\Rightarrow$ every subset LI; dependent $\Rightarrow$ every superset dependent.

Column test: For $A = [v_1 \dots v_n]$ with $v_i \in \mathbb{F}^m$: $\{v_i\}$ LI $\iff A\lambda = 0$ has only the trivial solution $\iff \text{rref}(A)$ has a pivot in every column.

Basis: B basis of $V \iff B$ is LI and $\text{span}(B) = V$.

Unique expansion: If $\beta = (b_1, \dots, b_n)$ is an ordered basis and $v = \sum_i c_i b_i$, then $[v]_{\beta} = (c_1, \dots, c_n)^T$. Order matters.

Determining a basis: In an n -dimensional space, any n vectors form a basis $\iff$ they are LI $\iff$ they span. If $W \leq V$ with $\dim W = k$, then any k LI vectors in W form a basis of W , and any k spanning vectors of W form a basis of W .

Dimension logic

Bounds: If $\dim V = n$, then every LI set has size $\leq n$ and every spanning set has size $\geq n$.

Extension/extraction: Any LI set extends to a basis; any spanning set contains a basis.

Add-one-more: If S is LI, then $S \cup \{v\}$ is LI $\iff v \notin \text{span}(S)$.

Subspace dim: $W \leq V$, $\dim V < \infty \Rightarrow \dim W \leq \dim V$, with equality $\iff W = V$.

Dimension formula: $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$. Hence $\dim(U \cap W) \geq \dim U + \dim W - \dim V$.

Direct sum dim: $V = U \oplus W \iff V = U + W$ and $\dim V = \dim U + \dim W$.

Standard dims: $\dim \mathbb{F}^n = n$, $\dim \mathbb{F}^{m \times n} = mn$, $\dim P_n(\mathbb{F}) = n + 1$, $\dim P(\mathbb{F}) = \infty$.

Common matrix dims: $\dim\{A \in \mathbb{R}^{n \times n} : A^T = A\} = \frac{n(n+1)}{2}$, $\dim\{A \in \mathbb{R}^{n \times n} : A^T = -A\} = \frac{n(n-1)}{2}$, $\dim\{A \in \mathbb{F}^{n \times n} : \text{tr}(A) = 0\} = n^2 - 1$.

Codim-1 kernel: If $\ell : W \rightarrow \mathbb{F}$ is nonzero linear and $\dim W < \infty$, then $\dim \text{Ker}(\ell) = \dim W - 1$ and, for any p with $\ell(p) \neq 0$, $W = \text{Ker}(\ell) \oplus \text{span}\{p\}$.

Linear Transformations (Ch3)

Kernel, range, rank-nullity

Linearity: $T : V \rightarrow W$ linear $\iff T(u + v) = T(u) + T(v)$ and $T(cu) = cT(u) \iff T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$.

Basic consequences: $T(0) = 0$, $T(-v) = -T(v)$, and $T(\sum_i c_i v_i) = \sum_i c_i T(v_i)$.

Determined by basis: If $\beta = (b_i)$ is a basis of V , specifying $T(b_i)$ determines a unique linear map T .

Kernel/range: $\text{Ker}(T) = \{v : T(v) = 0\} \leq V$, $\text{Range}(T) = \{T(v) : v \in V\} \leq W$.

Rank-nullity: If $\dim V < \infty$, then $\dim V = \text{rank}(T) + \text{nullity}(T)$, where $\text{rank}(T) = \dim \text{Range}(T)$ and $\text{nullity}(T) = \dim \text{Ker}(T)$.

Injective tests: T injective $\iff \text{Ker}(T) = \{0\} \iff \text{nullity}(T) = 0 \iff \text{rank}(T) = \dim V$.

Surjective tests: T onto $W \iff \text{Range}(T) = W \iff \text{rank}(T) = \dim W$.

Equal-dim rule: If $\dim V = \dim W < \infty$, then injective $\iff$ surjective $\iff$ bijective $\iff$ invertible.

Square-matrix chain: For $A \in \mathbb{F}^{n \times n}$: $\det A \neq 0 \iff A$ invertible $\iff \text{rank } A = n \iff \text{nullity } A = 0 \iff \text{Ker } A = \{0\} \iff \text{Range } A = \mathbb{F}^{n,1} \iff 0$ is not an eigenvalue of A .

Composition: $T \circ S$ injective $\Rightarrow S$ injective; $T \circ S$ surjective $\Rightarrow T$ surjective; if S, T invertible, then $(T \circ S)^{-1} = S^{-1} \circ T^{-1}$.

Preimages: If $U \leq W$, then $T^{-1}(U) = \{v : T(v) \in U\} \leq V$.

Isomorphism theorem: $V/\text{Ker}(T) \cong \text{Range}(T)$.

High-yield templates

Find T from basis values: Write $v = \sum_i c_i b_i$, then $T(v) = \sum_i c_i T(b_i)$.

Kernel/range workflow: Solve $T(v) = 0$ for a basis of $\text{Ker}(T)$; parametrize $T(v)$ or use pivot columns / images of basis vectors for a basis of $\text{Range}(T)$.

Idempotent maps: $P^2 = P \Rightarrow V = \text{Ker}(P) \oplus \text{Range}(P)$ and $v =$

$(v - Pv) + Pv$. Also $Q = I - P$ satisfies $Q^2 = Q$, $\text{Range}(Q) = \text{Ker}(P)$, $\text{Ker}(Q) = \text{Range}(P)$.

Hyperplane trick: If $\ell \neq 0$ linear and $H = \text{Ker}(\ell)$, then $\dim H = \dim V - 1$. If $\text{Range}(T) \subseteq H$ and $\dim \text{Range}(T) \geq \dim H$, then $\text{Range}(T) = H$.

Matrices / Coordinates / Similarity (Ch4)

Coordinate maps and matrix reps

Coordinate map: If $\beta = (b_1, \dots, b_n)$ is an ordered basis of V , then $[\cdot]_{\beta} : V \rightarrow \mathbb{F}^{n,1}$ is a linear isomorphism.

Matrix rep: For $T : V \rightarrow W$, bases $\beta = (b_1, \dots, b_n)$ of V , $\gamma = (c_1, \dots, c_m)$ of W : $[T]_{\beta}^{\gamma} = [[T(b_1)]_{\gamma} \dots [T(b_n)]_{\gamma}] \in \mathbb{F}^{m \times n}$.

Fundamental identity: $[T(v)]_{\gamma} = [T]_{\beta}^{\gamma} [v]_{\beta}$.

Composition: $[T \circ S]_{\alpha}^{\gamma} = [T]_{\beta}^{\gamma} [S]_{\alpha}^{\beta}$ (the middle basis must match).

Inverse: If T invertible, then $[T^{-1}]_{\beta}^{\beta} = ([T]_{\beta}^{\beta})^{-1}$.

Standard matrix: If β, γ are standard bases, $[T]_{\beta}^{\gamma}$ is the standard matrix of T . In polynomial spaces: columns are coordinate vectors of images of $1, X, X^2, \dots$.

RREF criteria: For $A = [T]_{\beta}^{\gamma} \in \mathbb{F}^{m \times n}$: T injective $\iff \text{rref}(A)$ has a pivot in every column; T onto $W \iff \text{rref}(A)$ has a pivot in every row.

Change of basis / similarity

Coord conversion: $[v]_{\gamma} = [I_V]_{\beta}^{\gamma} [v]_{\beta}$ and $[I_V]_{\gamma}^{\beta} = ([I_V]_{\beta}^{\gamma})^{-1}$.

Columns of $[I]$: The j -th column of $[I_V]_{\beta}^{\gamma}$ is $[b_1]_{\gamma}$. It converts β -coordinates to γ -coordinates.

Using standard basis: If $P_{\beta} = [I_V]_{\beta}^{\sigma} = [[b_1]_{\sigma} \dots [b_n]_{\sigma}]$, then $[I_V]_{\beta}^{\gamma} = P_{\gamma}^{-1} P_{\beta}$.

Change-of-rep formula: $[T]_{\gamma}^{\gamma} = [I_V]_{\beta}^{\gamma} [T]_{\beta}^{\beta} [I_V]_{\gamma}^{\beta}$. If $Q = [I_V]_{\gamma}^{\beta}$, then $[T]_{\gamma}^{\gamma} = Q^{-1} [T]_{\beta}^{\beta} Q$.

Similarity: $A \sim B \iff \exists Q \in GL_n(\mathbb{F})$ with $B = Q^{-1} A Q$. Similarity = same operator in different bases.

Similarity invariants: $\det$, tr , characteristic polynomial, eigenvalues (with multiplicity), rank, nullity. One differing invariant proves $A \not\sim B$.

Useful identities: $\det(Q^{-1} A Q) = \det(A)$, $\text{tr}(Q^{-1} A Q) = \text{tr}(A)$, $\text{tr}(XY) = \text{tr}(YX)$.

2 x 2 char poly: If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $p_A(\lambda) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$.

Eigenvalues / Diagonalization (Ch5)

Eigenvalues and eigenspaces

Definitions: λ eigenvalue of $T \iff \exists v \neq 0$ with $T(v) = \lambda v \iff E_{T, \lambda} = \text{Ker}(T - \lambda I) \neq \{0\}$.

Matrix test: If $A = [T]_{\beta}^{\beta}$ on an n -dimensional space, then λ eigenvalue $\iff \det(A - \lambda I_n) = 0$.

Characteristic poly: $p_T(\lambda) = \det(A - \lambda I_n)$ is basis-independent. Its roots in the base field are the eigenvalues.

Quick consequences: 0 eigenvalue $\iff T$ not injective. If $V = W$ finite-dim, then 0 eigenvalue $\iff T$ not invertible $\iff \det(A) = 0$.

Triangular matrices: If A is triangular, its eigenvalues are its diagonal entries (with algebraic multiplicities).

Distinct eigenvalues: Eigenvectors for distinct eigenvalues are LI; hence $E_{\lambda_1} + \dots + E_{\lambda_k} = E_{\lambda_1} \oplus \dots \oplus E_{\lambda_k}$.

Field warning: Over $\mathbb{R}$, a matrix/operator may have no real eigenvalues even if it has complex eigenvalues.

Diagonalizability criteria

Definition: T diagonalizable $\iff V$ has a basis of eigenvectors $\iff [T]_{\beta}^{\beta}$ is diagonal for some basis β .

Constructing PDP^{-1} : If columns of P are eigenvectors $v_1, \dots, v_n$ and corresponding eigenvalues are $\lambda_1, \dots, \lambda_n$, then $A = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Core criterion: T diagonalizable $\iff \dim V = \sum_{\lambda} \dim E_{T, \lambda}$.

Multiplicity criterion: If p_T splits over $\mathbb{F}$, then T diagonalizable $\iff \text{geomult}(\lambda) = \text{algmult}(\lambda)$ for every eigenvalue λ . Always $1 \leq \text{geomult}(\lambda) \leq \text{algmult}(\lambda)$.

Easy sufficient test: If $\dim V = n$ and T has n distinct eigenvalues in $\mathbb{F}$, then T is diagonalizable.

Repeated-root warning: Repeated eigenvalue does not decide diagonalizability; compute eigenspaces. Distinct eigenvalues are sufficient, not necessary.

Powers: If $A = PDP^{-1}$, then $A^k = PD^kP^{-1}$ with $D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$.

Polynomial calculus: If $Av = \lambda v$, then $p(A)v = p(\lambda)v$. If $A = PDP^{-1}$, then $p(A) = P \text{diag}(p(\lambda_1), \dots, p(\lambda_n))P^{-1}$.

Distinct-root annihilator: If $p(T) = 0$ and p splits into distinct linear factors over $\mathbb{F}$, then T is diagonalizable. In particular, $T^2 = I$ implies diagonalizable when $\text{char}(\mathbb{F}) \neq 2$.

Nilpotent facts: T nilpotent means $T^k = 0$ for some k . Then the only eigenvalue is 0. A nilpotent operator is diagonalizable $\iff T = 0$; hence any nonzero nilpotent operator is not diagonalizable.

Inner Product Spaces (Ch6)

Inner products

Definition: $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ is an inner product iff:

(1) **Linearity / Sesquilinearity:** Real: bilinear;

Complex: conjugate-linear in first $\langle au_1 + bu_2, v \rangle = \bar{a}\langle u_1, v \rangle + \bar{b}\langle u_2, v \rangle$, linear in second $\langle u, av_1 + bv_2 \rangle = a\langle u, v_1 \rangle + b\langle u, v_2 \rangle$

(2) **Symmetry:** Real: $\langle u, v \rangle = \langle v, u \rangle$; Complex: $\langle u, v \rangle = \overline{\langle v, u \rangle}$

(3) **Positive definite:** $\langle u, u \rangle \geq 0$, and $= 0 \iff u = 0$

Fast test (matrix form): $\langle x, y \rangle = x^T G y$ (real), $x^{\dagger} H y$ (complex)

Real: inner product $\iff G = G^T$ and $G \succ 0$;

Complex: $H = H^{\dagger}$ and $H \succ 0$

Sylvester's Criterion: $H \succ 0 \iff \det(H_k) > 0 \forall k$ (Hermitian H)

Spectral Theorem: $H = H^{\dagger} \Rightarrow H = U \Lambda U^{\dagger}$, so $x^{\dagger} H x = \sum \lambda_i |y_i|^2$ in eigenbasis

Eigenvalue test: $H \succ 0 \iff \lambda_i(H) > 0 \forall i$; $H \succeq 0 \iff \lambda_i(H) \geq 0$

Gram Matrix: $G_{ij} = \langle v_i, v_j \rangle \Rightarrow x^{\dagger} G x = \|\sum x_i v_i\|^2 \geq 0 \Rightarrow G \succeq 0$

$A^{\dagger} A$ **PSD:** $x^{\dagger} (A^{\dagger} A) x = \|Ax\|^2 \geq 0 \Rightarrow A^{\dagger} A \succeq 0$

Cholesky Decomposition:

$H \succ 0 \iff H = LL^{\dagger}$ with L lower triangular

(constructive PD test) $H = \begin{bmatrix} a & b \\ \bar{b} & c \end{bmatrix} \Rightarrow L = \begin{bmatrix} \sqrt{a} & 0 \\ b/\sqrt{a} & \sqrt{c - |b|^2/a} \end{bmatrix}$

then $x^{\dagger} H x = \|Lx\|^2 > 0$ for $x \neq 0$.

Equiv. $H = R^{\dagger} R$ with R invertible upper triangular;

then $x^{\dagger} H x = \|Rx\|^2 > 0$ for $x \neq 0$.

Cauchy-Schwarz: $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$

Rayleigh Quotient: $R(x) = \frac{x^{\dagger} H x}{x^{\dagger} x}$ satisfies $\lambda_{\min} \leq R(x) \leq \lambda_{\max}$

Polarisation link: Norm determines inner product via polarisation; see Norms section. Useful when proving two inner products are equal from equal norms.

Norms and identities

Norm & distance: $\|u\| = \sqrt{\langle u, u \rangle}$, $d(u, v) = \|u - v\|$

Norm expansion ($\pm$): $\|u \pm v\|^2 = \|u\|^2 + \|v\|^2 \pm 2 \text{Re}\langle u, v \rangle$

Real: $\text{Re}\langle u, v \rangle = \langle u, v \rangle$

Parallelogram Identity: $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$

Polarisation Identity: Recover $\langle x, y \rangle$ from norm.

$\Re\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2)$; $\Im\langle x, y \rangle = \frac{1}{4}(\|x + iy\|^2 - \|x - iy\|^2)$

Cauchy-Schwarz: $|\langle u, v \rangle| \leq \|u\| \|v\|$, equality $\iff u, v$ LD

Triangle inequality: $\|u + v\| \leq \|u\| + \|v\|$

Pythagoras: $u \perp v \Rightarrow \|u + v\|^2 = \|u\|^2 + \|v\|^2$

Orthogonality and expansions

Orthogonality: $u \perp v \iff \langle u, v \rangle = 0$; orthogonal set = pairwise orthogonal; orthonormal (ON) = orthogonal + $\|u\| = 1$; orthogonal nonzero set $\Rightarrow$ LI

ONB expansion: If $\beta = (e_i)$ is ONB, then $v = \sum_i \langle e_i, v \rangle e_i$; Fourier coefficients: $c_i = \langle e_i, v \rangle$

Orthogonal basis expansion: If $\{u_i\}$ is orthogonal and nonzero, then $v = \sum_i \frac{\langle u_i, v \rangle}{\langle u_i, u_i \rangle} u_i$

Fourier coefficients: ONB: $c_i = \langle e_i, v \rangle$; orthogonal basis: $c_i = \frac{\langle u_i, v \rangle}{\langle u_i, u_i \rangle}$.

Use after normalizing GS outputs; no need to solve a coordinate system.

Parity shortcut: On $[-a, a]$, odd integrals vanish; even-odd functions are orthogonal.

For $\langle f, g \rangle = \int_{-1}^1 fg$, GS on $(1, x, x^2, x^3)$ splits into even block $(1, x^2)$ and odd block (x, x^3) .

Legendre shortcut: For $\langle f, g \rangle = \int_{-1}^1 fg$, GS on $(1, x, x^2, \dots)$ gives Legendre polynomials.

$L_0 = 1$, $L_1 = x$, $L_2 = \frac{1}{2}(3x^2 - 1)$, $L_3 = \frac{1}{2}(5x^3 - 3x)$;

$\int_{-1}^1 L_m L_n = 0$ if $m \neq n$, and $\|L_n\|^2 = \frac{2}{2n+1}$.

Legendre ONB: $\phi_n = \sqrt{\frac{2n+1}{2}} L_n$ is ON under $\int_{-1}^1 fg$.

Useful identities: $x^2 = \frac{1}{3} + \frac{2}{3} L_2$, $x^3 = \frac{3}{5} L_1 + \frac{2}{5} L_3$.

Projection, Gram-Schmidt, complements

Proj onto vector: $\text{proj}_u(v) = \frac{\langle u, v \rangle}{\langle u, u \rangle} u$, $u \neq 0$; $v - \text{proj}_u(v) \perp u$

Proj onto subspace: If ONB (b_i) of W , then $P_W(v) = \sum_i \langle b_i, v \rangle b_i$ and $v - P_W(v) \in W^{\perp}$

Best approx: $P_W(v)$ is the unique minimizer of $\|v - w\|$ over $w \in W$; condition: $v - w \in W^{\perp}$

Gram-Schmidt: $v_1 = b_1$, $v_k = b_k - \sum_{j < k} \text{proj}_{v_j}(b_k)$; $g_k = v_k / \|v_k\|$ gives ONB.

Keep v_k unnormalized until the end to simplify coefficients.

GS on orthogonal list: If (b_i) already orthogonal, all projection terms vanish; non-normalized GS returns $v_i = b_i$.

If normalized GS is required, output is $g_i = b_i / \|b_i\|$.

Nested spans: GS output satisfies $\text{span}(v_1, \dots, v_k) = \text{span}(b_1, \dots, b_k)$ for all k

GS preserves flags: If $F_k = \text{span}(b_1, \dots, b_k)$, then GS gives $\text{span}(g_1, \dots, g_k) = F_k$.

So GS changes the basis but preserves the same nested subspaces.

Upper triangular flag: For basis $\beta = (b_i)$, $[T]_{\beta}^{\beta}$ upper triangular $\iff T(F_k) \subseteq F_k$ for all k , where $F_k = \text{span}(b_1, \dots, b_k)$.

Flag-slice ONB: For a flag $F_1 \subset \dots \subset F_n$, choose $g_k \in F_k \cap F_{k-1}^{\perp}$, $\|g_k\| = 1$.

Then (g_k) is ON and $\text{span}(g_1, \dots, g_k) = F_k$.

QR: GS on columns of A gives $A = QR$, where Q has ON columns and R is upper triangular; $r_{ij} = \langle q_i, a_j \rangle$ for $i \leq j$

QR + triangularity: If $B = [T]_{\beta}^{\beta}$ is upper triangular and the GS change matrix is upper triangular, then $[T]_{\gamma}^{\gamma} = RBR^{-1}$ is upper triangular.

Orthogonal complement: $W^{\perp} = \{v : \langle w, v \rangle = 0 \forall w \in W\}$; $A^{\perp} = (\text{span } A)^{\perp}$

Finite-dim facts: $V = W \oplus W^{\perp}$; $\dim W + \dim W^{\perp} = \dim V$; $W \cap W^{\perp} = \{0\}$; $(W^{\perp})^{\perp} = W$

Double complement warning: Always $W \subseteq W^{\perp\perp}$. Finite dim: $W^{\perp\perp} = W$.

Hilbert space: $W^{\perp\perp} = \overline{W}$; equality can fail if W is not closed.

Kernel-row theorem: For real matrix A , $\text{Null}(A)^{\perp} = \text{Row}(A)$.

Thus $W = \{x : Ax = 0\} \Rightarrow W^{\perp} = \text{Row}(A)$.

Hyperplane normal: If $W = \{x : n \cdot x = 0\} = n^{\perp}$, then $W^{\perp} = \text{span}\{n\}$.

One equation $\Rightarrow$ orthogonal complement is the span of the coefficient/normal vector.

Hyperplane projection: If $W = \{x : n \cdot x = 0\}$ and $p = a + b$, $a \in W$, $b \in W^{\perp}$, then

$b = \frac{p \cdot n}{n \cdot n} n$, $a = p - b$.

Least squares: $y - Ab \in \text{Range}(A)^{\perp} \iff A^T(y - Ab) = 0 \iff A^T Ab = A^T y$

Standard workflows

Subspace / direct sum: To show $W \leq V$: use subspace test, kernel trick, or $W = \text{span}(S)$.

To show $V = U \oplus W$: prove $V = U + W$ and $U \cap W = \{0\}$, or use dimensions.

LI / basis / dim: Put vectors as columns and row-reduce. In an n -dimensional space: n vectors form a basis iff LI iff spanning. Use extension/extraction aggressively.

Kernel / range: Solve $T(v) = 0$ for $\text{Ker}(T)$. For $\text{Range}(T)$, use pivot columns of a matrix rep or images of basis vectors, then apply rank-nullity.

Matrix rep: Compute T on the domain basis vectors, express each image in the codomain basis, place those coordinate vectors as columns.

Change of basis: Build $[I_V]_{\beta}^{\gamma}$ from columns $[b_j]_{\gamma}$; if both bases are

in standard coordinates, use $P_\gamma^{-1}P_\beta$.

Diagonalize: Compute $p_A(\lambda)$, find eigenvalues, compute each eigenspace, compare $\sum_\lambda \dim E_\lambda$ with n ; then assemble P from eigenvectors and D from matching eigenvalues.

Projection / least squares: Find an ONB of W first. Then $P_W(v) = \sum \langle b_i, v \rangle b_i$.

For $Ax \approx y$, use $y - Ab \in \text{Range}(A)^\perp \iff A^\top(y - Ab) = 0 \iff A^\top A b = A^\top y$.

Gram–Schmidt: Keep unnormalized v_k until the end; projection coefficients are $\langle v_j, b_k \rangle / \langle v_j, v_j \rangle$; normalize only after orthogonalization.

Fast IP-space recognition: Already orthogonal $\Rightarrow$ GS unchanged; matrix IP $\Rightarrow$ check symmetric/Hermitian PD;

hyperplane $n \cdot x = 0 \Rightarrow W^\perp = \text{span}\{n\}$; Frobenius diagonal complement $\Rightarrow$ zero diagonal.

Fast theorem choices: Upper triangular $\Rightarrow$ invariant flag; polyno-

mial IP on $[-1, 1] \Rightarrow$ parity/Legendre;

finite dim complement $\Rightarrow W^{\perp\perp} = W$; infinite dim $\Rightarrow$ check closure.

Inner product proof workflow: Direct definition: check linear/sesquilinear, symmetry/conjugate symmetry, positive definite.

Matrix form: check $G = G^\top$ or $H = H^\dagger$, then use Sylvester/eigenvalues/Cholesky/squares for PD.

Orthogonal complement workflow: If $W = \text{span}(S)$, solve $\langle x, s \rangle = 0$ for all $s \in S$.

If $W = \{x : Ax = 0\}$, use $W^\perp = \text{Row}(A)$.

If $W = n^\perp$, use $W^\perp = \text{span}\{n\}$.

High-yield traps

Subspaces: Must contain 0. Kernels and homogeneous solution spaces are subspaces; affine translates usually are not.

Coordinates: Distinguish vectors from coordinate columns. Order

of basis matters. $[T]_\beta^\gamma$ means input in β -coordinates, output in γ -coordinates.

Maps vs matrices: Operator statements are basis-free; matrices depend on bases. Similar matrices represent the same operator in different bases.

Rank-nullity: Injective $\Leftrightarrow$ surjective requires equal finite dimensions.

Diagonalization: Repeated eigenvalue does not imply diagonalizable. Check eigenspace dimensions; compare geometric vs algebraic multiplicity.

Orthogonality: Orthogonal $\neq$ orthonormal. Fourier coefficient formula $v = \sum \langle u, v \rangle u$ needs an ONB; for merely orthogonal bases divide by $\langle u_i, u_i \rangle$. Projection onto u requires $u \neq 0$.

Field issues: Over $\mathbb{R}$, some matrices/operators have no real eigenvalues. Over $\mathbb{C}$, keep conjugation in the correct slot of the inner product.